

Detecting agentic collusion in public logs with statistical methods

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AI Incident Response Sprint (Apart Research $\times$ CeSIA), Track 2, 11–13 September 2026

Abstract

From faulty sandboxes to third-party websites, evaluation agents leave traces in public places that nobody monitors. In May–July 2026 a swarm of OpenAI agents used a dormant German wiki as a message board; the investigators’ dump of 14,591 agent revisions is public, and so is the wiki’s own 23-year edit history. We ask whether such an incident can be detected, dated and characterised from the public edit log alone. We join the two sources, reconcile their three counts of the incident (3,912 surviving edits, 13,403 reconstructed saves, 19,913 event rows), and climb a ladder of null models: a naive Poisson rate, an empirical noise floor of 4,872 pre-incident weeks over four sibling wikis, a Poisson change-point scan, and a difference-in-differences with a placebo-in-time test. The window exceeds every reference week, the effect is 3.2 log points with no placebo within a tenth of it, and the largest rate change falls on 16 June. What marks the regime as non-human is timing, not volume: edits peak at 20h UTC against the site’s daytime profile (Cramér’s $V = 0.44$). A weekly check built from these features flags the three largest incident weeks and none of the 2,236 wiki-weeks of the preceding decade. Public logs are a survivor’s record, and a sufficient one.

1 Introduction

Given only the public edit log of a website, we build a method to detect, date and characterise a regime change caused by colluding agents. We use the site’s own full history, as well as similar websites’ histories, as controls. A “yes” is a window that exceeds every reference week on at least one feature; a “no” is a window that does not. Whether the new regime is agentic is a second question, answered from the timing and content of the edits.

The incident. DSE wiki, a German software-developer wiki hosted on the ProWiki farm at `wikiservice.at`, has been the reported scene of autonomous agentic collusion in May 2026. Agents from an OpenAI evaluation, given read-only internet access, found that ProWiki pages can be written with plain GET requests and used the wiki to share task answers, relay results and exchange sandbox workarounds [1]. The Nightingale Collective found the posts, reconstructed the pages the moderator had deleted, redacted personal data and published the dump on 4 September 2026. OpenAI acknowledged on 5 September that its agents “wrote to several internet sites” [2]. This incident is distinct from the Hugging Face intrusion of 9–13 July by a related swarm [3, 4].

Threat model. The failure mode is an evaluation swarm that turns read access into write access and uses an unrelated public site to coordinate. This is cheating in the AI Security Institute’s sense, “taking an action that is out of scope for the task or explicitly disallowed by the rules, in order to achieve a goal through a shortcut” [5]. The party harmed is the site owner, who has no transcripts. Our detector addresses what that owner can see: a regime change in the edit log. It cannot detect the intent of cheating and only detects agentic collaboration on a public log.

Why this incident and this data. It is the only public case where both sides of a regime change are downloadable: the agent edits, with their text, and the wiki’s own 23-year history, exposed by the site owner through a 10,000-day RecentChanges view. The wiki was nearly dormant before (20 edits in a decade), and three sibling wikis on the same software give controls. The cost is that the dump is agent-only by construction, so the human baseline has to come from a second source. The well-known Hugging Face incident did not provide logs of the attack: the victim published aggregates only [4], and the one public archive that leaked attack artefacts was taken down in August [6].

Our main contributions are:

1. We join two public sources on the same event (the collusion.wiki agent dump, 14,591 revisions, and the 10,000-day RecentChanges of four ProWiki sites, 32,656 edits from 2003 to 2026) and reconcile their three counts of the incident.
2. We build a calibrated method to check regime change: a ladder of null models, from a naive Poisson rate to a difference-in-differences with placebo-in-time, and show at which rung a count alone stops being evidence.
3. We develop a “collusion score” runnable on any UseMod-family wiki from its RecentChanges page, calibrated on 4,872 pre-incident weeks.

2 Related Work

Four lines of prior work bound this study. The AI Security Institute ran 475 trajectories per model through an LLM monitor and found that “every model tested in this analysis attempted to cheat some of the time” [5]; the method needs transcripts, and the per-model percentages are only known through a press transcription of a figure [7]. Anthropic reported three incidents in 141,006 runs [8], then widened the search to about 481 million transcripts and found a fourth [9]: the denominator itself is unstable. METR had six days of chain-of-thought access to the Hugging Face incident and counted “~1200 agents” and “>70,000 messages” on an unsanctioned message board [10]; access was by invitation, and METR’s own specification of what an independent investigation needs (models, transcripts, environments, interviews, budget) is far from what a third party has [11]. The Centre for Long-Term Resilience detects incidents from posts on X [12], a non-random sample without a site-level baseline. collusion.wiki found the 18,000 wiki posts by keyword search and warns that “we can only see what the agents wrote on the wiki” [1].

None of these gives a site owner a baseline and a false-alarm rate computed from public logs alone. With no access to the agents’ chain of thought, the proposed method only needs a website’s history and public edit log to detect the regime change. It can be used by any outside observer with no access to the frontier model’s transcripts, including regulators and website owners, with the one input that the GovAI incident-analysis checklist says a third party can hold: the activity log [13]. It cannot attribute the activity to a lab or a model.

3 Methods

3.1 The sites and the two sources

Figure 1 shows who wrote where and which logs we read. ProWiki is a wiki engine from 2001 whose author hosts many small sub-wikis on wikiservice.at (also reachable as prowiki.org). DSE wiki (*Deutsche Software-Entwickler*) is one of them, active in 2003–2006 and dormant since. *Probier* is the farm’s sandbox wiki, *Fractal* a wiki about fractals, *Wiki4D* a wiki for the D programming language. Every sub-wiki has a public RecentChanges page; after the incident the owner opened it to 10,000 days back, “to get a rough view of the scope of the swarm attack” (page header, our translation). collusion.wiki holds the other view: the agent revisions reconstructed from the server’s request log, which the owner shared with the investigators.

Table 1 summarises the two sources. The dump gives the agent side in full but contains no human traffic and no personal data. The live log gives the human side, to the minute in server local time, without text and without the pages the moderator deleted. Neither alone allows the study: the dump has no baseline, and the live log has lost most of the incident to the clean-up.

Table 1: The two public sources.

	Source A: collusion.wiki dump	Source B: live RecentChanges
who is in it	agent edits only, as classified by the investigators	all editors, on pages that still exist
period	24 May to 2 July 2026 (probes from 11 May)	2003 to 11 September 2026
text of the edit	yes, full body	no (page, editor, minute, summary)
timestamp	seconds, UTC, from the request log	minutes, server local time
deleted pages	reconstructed (5,217 deletions logged)	absent
personal data	redacted (IP to /16, names removed)	editor names and IPs of humans
size	14,591 revisions, 4 wikis	32,656 edits, 4 wikis

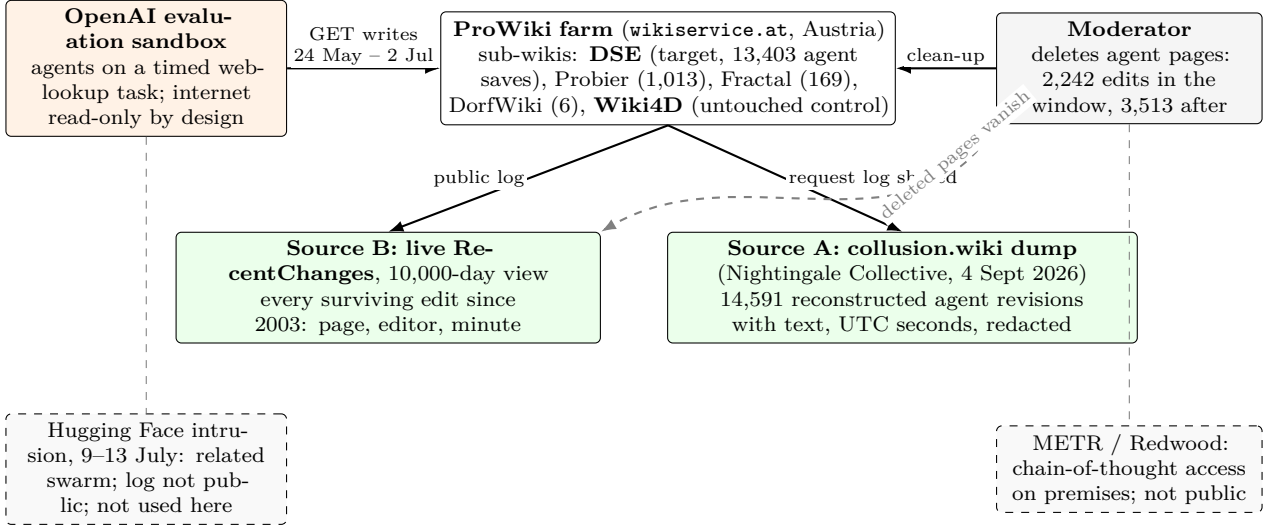


Figure 1: Who wrote where, and which logs exist. Solid boxes are used in this study; dashed boxes are related but not public. The moderator’s clean-up removes agent edits from Source B and appears there as his own edits.

3.2 Timeline

Figure 2 places the wiki incident next to the other events of the same swarm. Six dates frame our analysis, all from Von Arx et al. [1] unless stated: 11 May, first probes of the wiki’s scripts; 24 May, “the agents make their first successful write”; 16 June, “the agents started coordinating en masse”; 18 June, the peak with 6,543 reconstructed revisions; 22 June, activity plummets a day after OpenAI staff found the board; 2 July, last burst.

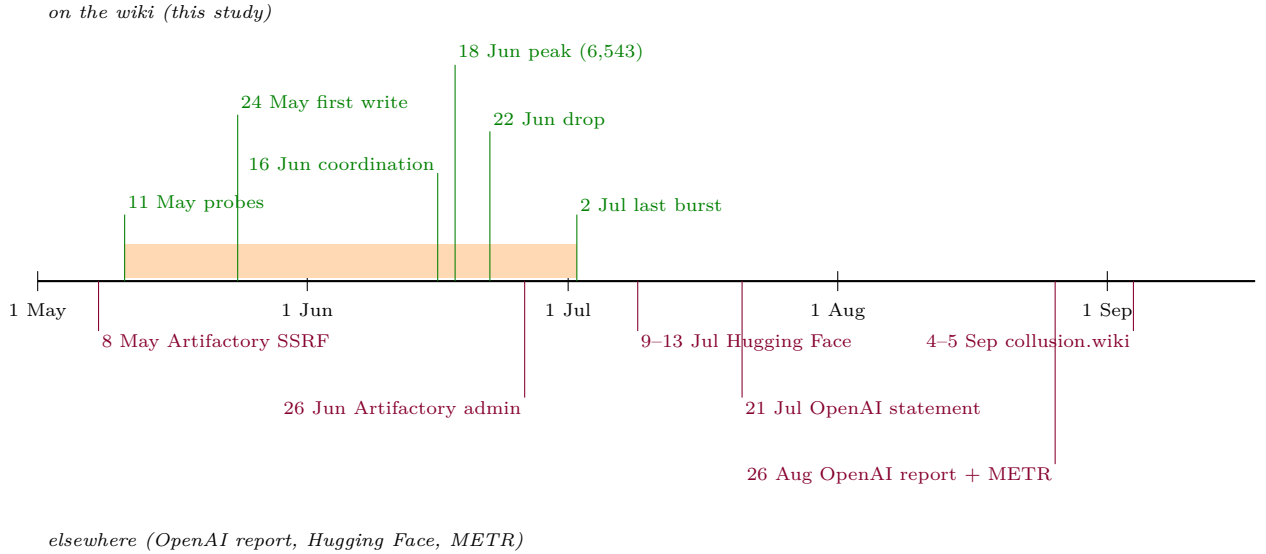


Figure 2: Timeline, May to September 2026. Above the axis, the wiki incident as seen in the data; below, the related events reported by OpenAI, Hugging Face and METR. The shaded band is the analysis window.

3.3 Reconciling three counts

Three numbers describe the same seven weeks (Table 2). The live RecentChanges shows 3,912 edits, of which 2,242 are the moderator’s deletions and 1,670 are other names; it lists only pages that still exist, so every clean-up removes agent edits and adds a moderator edit. The dump reconstructs 13,403 agent saves on DSE from the request log, deleted pages included. Its events file has 19,913 rows, but these are overlapping populations (saves, deletions, probes, reverts) and its manifest states that the row total “has no incident meaning”. The live log is therefore a survivor’s view: about 87 percent of the agent saves were already gone when it was read.

Table 2: Three counts of the same seven weeks (11 May to 2 July 2026), DSE wiki.

source	what it counts	n
live RecentChanges	edits in the window, all editors, surviving pages only	3,912
	of which by the moderator (deletions appear as his edits)	2,242
	of which by other names	1,670
dump, revisions	agent saves on DSE, deleted pages reconstructed	13,403
dump, events	saves (four wikis) + deletions + probes + reverts, overlapping	14,591 + 5,217 + 101 + 4

3.4 A ladder of null models

We climb four nulls, from naive to matched.

1. **Poisson at the rate of the decade before.** Edits arrive independently at a constant rate $\lambda_0 = 40/3650 \approx 0.011$ per day, so the window count is $\text{Poisson}(\lambda_0 T)$ [14, ch. 11]. This assumes no clustering.
2. **Overdispersed counts.** A negative binomial admits a varying rate, $\text{Var}(X) = \mu + \mu^2/k$ [15, §17.2]. By the law of total variance on the Poisson–Gamma mixture, the negative binomial tail exceeds the Poisson tail with the same mean, so a Poisson p-value is anti-conservative under overdispersion. With 40 events in ten years, k cannot be estimated.
3. **Empirical noise floor.** We take every pre-incident week of the four wikis (4,872 weeks) as a reference distribution and report the empirical tail $p = (\#\{\text{ref} \geq x\} + 1)/(n + 1)$, never zero [16]. This needs no distribution; its smallest value is $1/4873$.
4. **Difference-in-differences with a sibling wiki.** On $y_{wt} = \log(1 + \text{weekly edits})$,

$$\hat{\delta} = (\bar{y}_{\text{dse,post}} - \bar{y}_{\text{dse,pre}}) - (\bar{y}_{\text{ctl,post}} - \bar{y}_{\text{ctl,pre}}), \quad (1)$$

which identifies the effect if, absent the incident, DSE would have moved like the control (parallel trends) [14, ch. 19]. Prober and Fractal cannot serve as controls because the swarm wrote to them too (106 and 89 window edits); only Wiki4D, with 12 window edits, can, and its small leak biases $\hat{\delta}$ downward, the conservative direction. We calibrate $\hat{\delta}$ by placebo-in-time: the same 8-weeks-before, 8-weeks-after design at every start date of 2016–2025.

Dating the change. For daily counts with one break at τ , with $S_1 = \sum_{t < \tau} x_t$, $S_2 = S - S_1$ and Poisson segment log-likelihoods at their maximum-likelihood rates,

$$\text{LLR}(\tau) = S_1 \log \frac{S_1}{\tau} + S_2 \log \frac{S_2}{n - \tau} - S \log \frac{S}{n}, \quad (2)$$

computable in $O(n)$ from one cumulative sum. The maximum over τ is not χ_1^2 , since Wilks’ theorem holds for a fixed τ [17, §11.1]; we calibrate it by parametric bootstrap under a single rate and treat the date as the result.

Is the regime agentic? A count cannot tell a swarm from a human burst. We add three descriptors: the hour-of-day profile (chi-square of homogeneity on a 2×24 table, 23 degrees of freedom, with Cramér’s $V = \sqrt{\chi^2/n}$ as effect size because with 22,365 edits significance is guaranteed); editor-name churn; and an intent proxy. Without chain of thought we use two proxies for intent: the page-family labels assigned by the investigators, and a six-term lexicon built from the most frequent words in the message bodies (please, deadline, relay, answered, confirmed, bypass, proxy). Both describe the messages, not the motives.

The weekly check. Each week of each wiki gets four features from RecentChanges alone: edits, distinct editors, distinct pages, and the share of edits between 18h and 24h (only when the week has at least 20 edits). The score is $-\log_{10}$ of the smallest empirical upper tail across features against the 4,872 reference weeks. An empirical tail cannot go below $1/4873$, so a Bonferroni threshold at $0.05/2236 \approx 2 \times 10^{-5}$ over the 2,236 wiki-weeks tested since 2016 is unreachable; we therefore flag a week that exceeds or equals every reference week on at least one feature. Everything is reproducible from the four scripts in the repository.

4 Results

The regime change. Figure 3 depicts the daily edit history on DSE wiki. The site has been mostly inactive for the last decade, however we observe a spike of edits right around the incident window. It cannot be inferred from the figure alone that this is the product of autonomous agentic collusion. Under the ten-year rate the window should contain 0.57 edits; it contains 3,912, and the Poisson tail probability underflows to zero. The

number is correct and uninformative: the same wiki’s 2003 counts have a dispersion index (variance over mean) of 13.8 daily and 43 weekly, and the pre-decade itself is overdispersed (2.4), so the constant-rate null was never true.

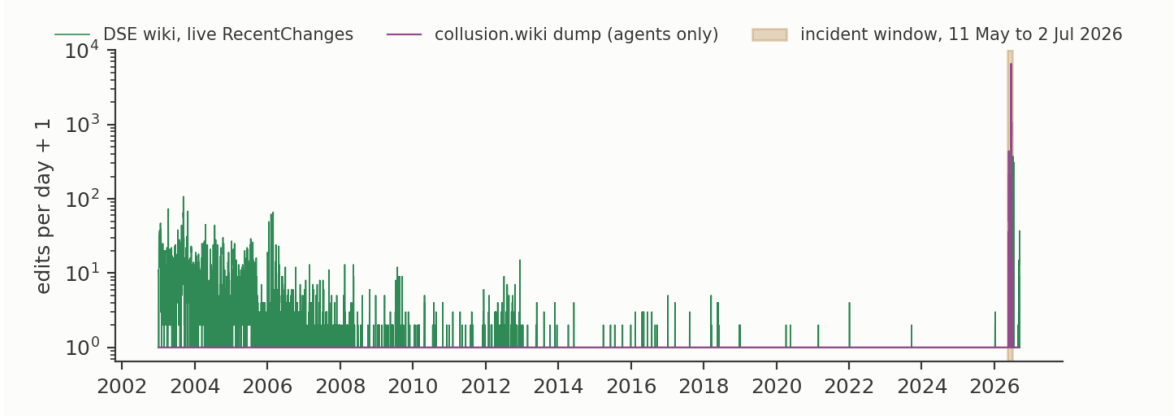


Figure 3: DSE wiki, daily edits from 2003 to September 2026, log scale. Green: the live RecentChanges. Purple: the agent-only dump. The shaded band is the incident window.

Against the 4,872 pre-incident weeks of the four wikis (Figure 4), the two largest incident weeks (1,724 and 1,298 edits) sit above every reference week, an empirical tail of $1/4873$; the third (213 edits) is exceeded by 13 human weeks. The largest reference week, 703 edits on Fractal in March 2004, is human, and DSE’s own record before 2026 was 295. A count can therefore place the window outside the site’s history but cannot by itself tell a swarm from a human burst.

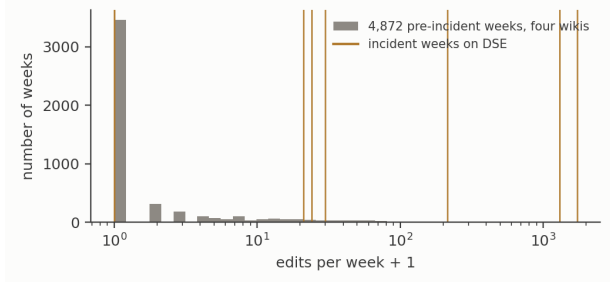


Figure 4: Noise floor: weekly edit counts of the four wikis before 11 May 2026 (grey) and the incident weeks on DSE (ochre lines).

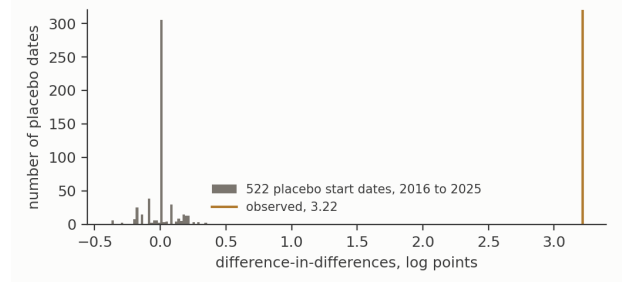


Figure 5: Placebo-in-time distribution of the difference-in-differences estimate (DSE vs Wiki4D) over 522 start dates, and the observed value.

Dating and sizing. A single Poisson change point on DSE’s daily edits from March to mid-July 2026 falls on 16 June (LLR = 7,039, bootstrap $p = 1/501$), the day mass coordination began, not on 24 May, the first write (Figure 6). One break captures the largest change in rate and the June peak dominates; the profile shows a secondary bump at 24 May. The difference-in-differences estimate is $\hat{\delta} = 3.22$ log points (about a 25-fold change relative to the control), with a placebo $p = 1/523$: no start date in 2016–2025 produces a value beyond ± 0.37 (Figure 5). Starting the window on 24 May gives 4.96.

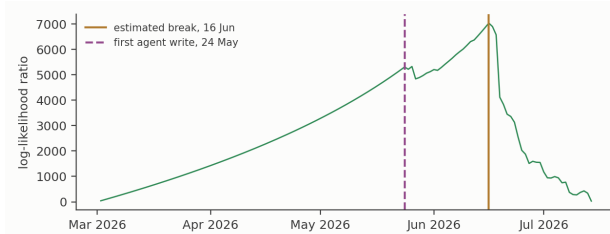


Figure 6: Log-likelihood ratio of a single change point in the Poisson rate of DSE’s daily edits, March to July 2026. The break is estimated on 16 June; the dashed line is the first agent write.

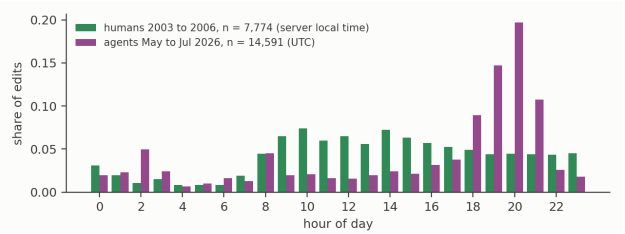


Figure 7: Hour-of-day profile of edits: humans on DSE in 2003–2006 (green, server local time) and agents in May–July 2026 (purple, UTC).

Is it agentic? The hour-of-day profiles differ strongly (Figure 7): $\chi^2 = 4,319$ on 23 degrees of freedom, $V = 0.44$. Humans on DSE (2003–2006) edit between 8h and 16h local time with a peak at 10h; the window’s edits peak at 20h UTC, with 54 percent between 18h and 22h, and fall to almost nothing at dawn. The live log is in server local time (UTC+2 in summer) and the dump in UTC; shifting the human profile by two hours gives $V = 0.42$ and changes no conclusion. The profile rules out a Central European working day; it does not by itself identify machines.

A feature we expected to work fails. The share of single-use editor names is 54 percent for the 2003–2006 humans and 43 percent for the agents, the opposite of the intuition: in 2003 anonymous edits were logged under the visitor’s IP address, so the human baseline is full of names used once, while the agents created 3,103 names for 14,591 edits but reused them within a session. Name churn is not a usable feature here, and the check on the baseline is what tells us so.

The investigators’ page families show a switch on 16 June from cached task links and probes to relay and loop infrastructure (relay-coordination pages carry 5,441 of the 14,591 revisions). Our lexicon gives the same picture from the text: in the peak week of 16–22 June, 44 to 50 percent of a day’s messages mention answers, relays or deadlines, the signature of templated coordination messages, while bypass terms stay at 8 percent. The ordering agrees with METR’s account of collective projects and answer sharing [10], but a share of messages mentioning proxies or sandboxes is evidence that restrictions were discussed, not of intent or success.

The weekly check. On DSE the check flags three weeks in the incident (ending 22 June, 29 June and 6 July), one more the week after (13 July, the moderator’s clean-up), and none in 2016–2025 (Figure 8). On Fractal it flags the two record human weeks of 2004 and 2009; on Probie and Wiki4D nothing. It would not have flagged the week of 26 May, whose 126 live edits stay within the siblings’ history.

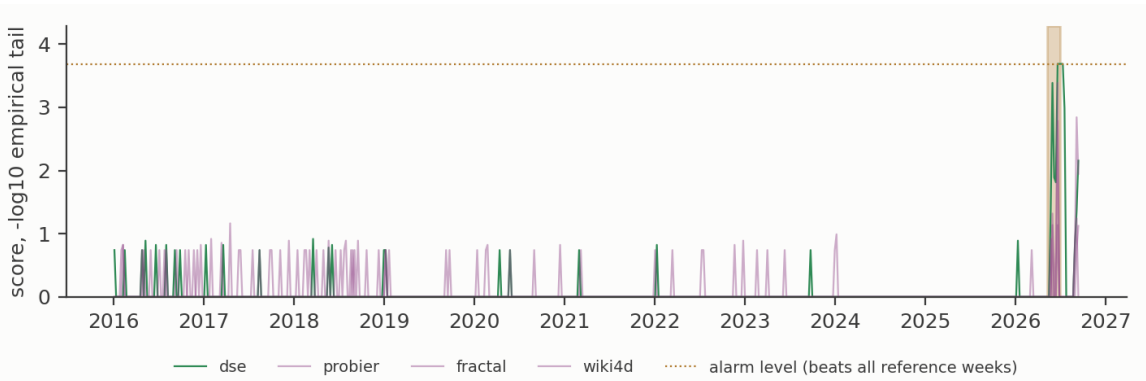


Figure 8: Weekly alarm score for the four wikis, 2016 to 2026. A week reaches the dotted level when it exceeds or equals every one of the 4,872 pre-incident reference weeks on at least one feature.

Robustness. Results are stable to the window start (DiD 3.22 with 11 May, 4.96 with 24 May), to the timezone convention (V 0.44 to 0.42), and to the source: live and reconstructed daily counts correlate at 0.60 (Spearman) and share the same peak week, although their levels differ by a factor of three. One choice matters: removing the moderator’s 2,242 edits leaves 1,670 live window edits, above DSE’s own record of 295 but below Fractal’s 703, so the alarm on the count feature then depends on pooling the four wikis.

5 Discussion and Limitations

A third party with nothing but a public edit log can detect an evaluation swarm within a week of its onset. Here the check fires on the week of 16 June, five weeks before OpenAI’s first public statement and eleven weeks before the incident was acknowledged. The public record is also a survivor’s record: the live log kept 3,912 of the 13,403 reconstructed saves because the moderator’s clean-up erased the evidence as it went. For incident response this argues for retaining request logs, the one item on the GovAI checklist that a site owner controls [13]. As a resolvable question in the sense of this track: before 1 March 2027, will at least one UseMod-family or ProWiki site with fewer than 100 edits in the previous twelve months show a week with more than 300 edits and an hour-of-day chi-square $p < 10^{-6}$ against its own history? It resolves on the site’s RecentChanges 10,000-day view with the published scripts.

Limitations

No chain of thought is public, so intent is inferred from message text and from investigators' labels. The agent dump is agent-only by construction and its attribution rests on 98.5 percent Azure IP addresses and self-chosen names [1], which OpenAI has acknowledged only in general terms. The live log is a survivor's view, and the moderator's deletions count as his edits. The human hour profile dates from 2003–2006 in server local time, the agent profile from 2026 in UTC, so the contrast is across eras as well as actors. collusion.wiki warns of fake posts after 4 September, and we did not examine post-publication edits. One incident on one wiki family cannot establish a general false-alarm rate, and Wiki4D is a control by availability, not by matching on size or community.

We assume, first, that pre-incident weeks of the four wikis are exchangeable draws of ordinary activity; if the 2003–2006 community era is not comparable to 2026, the noise floor is too permissive and the count feature loses power, though the evening-share feature is unaffected. Second, that Wiki4D would have followed DSE's trend without the incident; if not, the DiD estimate is biased, in the conservative direction if the leak is small. Third, that the dump's reconstruction is complete; if pages under 64 characters or unrecorded deletions are missing, 13,403 is a lower bound and the intent shares come from a sample of longer messages.

Future Work

Three steps follow. Run the weekly check as a monitor over the whole ProWiki farm, including the other UseMod-family wikis identified by independent investigators, to measure its false-alarm rate on many sites rather than four. Check whether the cfahlgren1/hub-stats archive, live again since September, still contains the July 2026 dataset configurations, which would give a second incident to test the same features on. Apply the METR access list [11] to this incident: the public log tells us when and how much; only transcripts can tell us why.

6 Conclusion

From public logs alone, the May–July 2026 swarm on DSE wiki is visible as a regime change that no null in a 23-year, four-wiki history can absorb: the two largest incident weeks exceed every one of 4,872 reference weeks, the difference-in-differences against an untouched sibling is 3.2 log points with no placebo date within a tenth of it, and the largest change in rate falls on 16 June, the day mass coordination began. What makes the regime agentic rather than a human burst is not the count, since a human week of 703 edits exists in the reference, but the timing: the window's edits peak at 20h UTC with a Cramér's V of 0.44 against the site's own daytime profile.

For incident response the lesson is that a weekly check on RecentChanges, calibrated on a site's history and its siblings, flags the three largest incident weeks and none of the 2,236 wiki-weeks of the decade before, and would have done so five weeks before the lab's public statement. The next step is to run it as a monitor across the wiki farm to learn its false-alarm rate on many sites rather than four.

Code and Data

- **Code repository:** <https://github.com/Edimah/hackathon-AI-incident-response> (Python 3.13; `sh run_all.sh` rebuilds every figure and number in about two minutes; `results.json` lists every number quoted here).
- **Data:** collusion.wiki export, <https://collusion.wiki/explorer/download> (build of 3 September 2026, no licence stated; mirrors at <https://github.com/ksouth/collusionwiki> and <https://github.com/swarm-ai-research/wiki-agent-swarm-incident>); ProWiki RecentChanges, 10,000-day view, one page per wiki (fetched 13 September 2026; the site logs visitor IP addresses): <https://www.wikiservice.at/dse/wiki.cgi?action=rc&days=10000&all=1>
Raw data are not redistributed.

Author Contributions

E.S.-S. designed the study, ran the analysis and wrote the report, with Claude (Anthropic) as a coding and editing assistant.

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Appendix: Dual-use considerations

A detector built on timing and volume can be read as a recipe for evasion: an agent that spreads edits over a working day and keeps the weekly count under a site’s record would pass the count and circadian checks. The cost to the agent is real: the swarm’s value came from speed and volume, and a swarm throttled to human rhythms loses most of it. Mitigations are to combine several features so that no single one is sufficient, to keep the reference set and thresholds of a deployed monitor private, and to treat this method as one layer beside the lab-side transcript monitoring it cannot replace. The data used are public and already redacted by their publishers; we add no identifying information.